

# SACHIN TENDULKAR

# CAREER ANALYSIS

## Cricket Data Analytics Project

Innings-level statistical analysis of a legendary 24-year international career

|            |           |         |               |             |             |
|------------|-----------|---------|---------------|-------------|-------------|
| 34,357     | 100       | 164     | 248           | 48.53       | 67.61       |
| Total Runs | Centuries | Fifties | Highest Score | Batting Avg | Strike Rate |

|              |                                                |
|--------------|------------------------------------------------|
| Project Type | Sports / Cricket Analytics                     |
| Domain       | Data Analysis — Python   SQL   Power BI        |
| Data Source  | ESPN Cricinfo (Innings-level career data)      |
| Career Span  | 1989 – 2013   782 Innings   74 Not Outs        |
| Tools Used   | Python (pandas, numpy) · SQL Server · Power BI |

## About the Dataset

This project analyses the complete innings-by-innings batting record of Sachin Tendulkar — widely regarded as the greatest batsman in cricket history. The dataset is sourced from ESPN Cricinfo and covers every international innings he played from November 1989 to November 2013.

| Attribute        | Detail                                                                |
|------------------|-----------------------------------------------------------------------|
| Dataset Type     | Sports / Cricket — Innings-level batting records                      |
| Nature           | Structured (Tabular)                                                  |
| Level            | Match & Innings level                                                 |
| Formats Covered  | Test, ODI, T20I                                                       |
| Numerical Cols   | Runs, Balls Faced, Fours, Sixes, Strike Rate, Mins, Position, Innings |
| Categorical Cols | Opponent, Ground, Match Type, Dismissal, Career Phase, Home/Away      |
| Date Range       | 15 Nov 1989 – 14 Nov 2013                                             |
| Total Records    | 782 innings (after cleaning)                                          |

## Domain Background

In international cricket, every batting innings is a structured data event. Performance analytics helps teams, selectors, and fans understand a player's consistency, adaptability, strengths against different opponents, and longevity. Sachin's career is the ideal dataset to demonstrate end-to-end data analytics because it is:

- Long (24 years) — enabling time-series and trend analysis
- Multi-format — Tests, ODIs, and T20Is offer comparative analysis
- Multi-geography — home vs away performance contrast
- Milestone-rich — 100 international centuries, 164 half-centuries

## Why This Data Is Important

Analysing Sachin's career data helps us:

- Identify peak performance years and career phases
- Understand format-specific batting behaviour (Test patience vs ODI aggression)
- Measure opposition difficulty using average and consistency metrics
- Track scoring rate (strike rate) evolution over 24 years
- Demonstrate real-world data cleaning, SQL querying, and BI dashboard skills

## Business Problem

How can we analyse Sachin Tendulkar's innings-level career data to extract meaningful performance insights — across formats, opponents, venues, and years — and present them through an interactive Power BI dashboard?

### Breaking Down the Problem

#### Career Overview Gap

No single view exists that combines total innings, runs, average, centuries, fifties, highest score, and strike rate into one dashboard.

#### Format Comparison

Fans and analysts want to know how Sachin's batting differed between Test matches (technique, patience) and ODIs (aggression, strike rate).

#### Opponent Difficulty Analysis

Which opponents challenged Sachin the most? Who were his favourite teams to score against?

#### Home vs Away Disparity

Did Sachin perform significantly better on home pitches (India) compared to overseas tours?

#### Year-on-Year Progression

How did his run tally, average, and strike rate evolve through Early Career, Prime, and Late Career phases?

#### Dismissal Pattern

Understanding his most common dismissal types reveals batting vulnerabilities.

#### Scoring Distribution

What percentage of innings ended in a duck, low score, fifty, or century? Is he consistent?

#### Conversion Rate

What percentage of fifties were converted into centuries — a key mental-strength indicator?

## Business Questions

| #   | Business Question                                                                         |
|-----|-------------------------------------------------------------------------------------------|
| Q1  | What is Sachin's total career innings count and total runs scored?                        |
| Q2  | What is Sachin's overall career batting average?                                          |
| Q3  | What does Sachin's complete career summary look like (all key stats in one view)?         |
| Q4  | How did Sachin perform across different match formats — Test, ODI, T20I?                  |
| Q5  | Which opponents did Sachin score the most runs against (min 10 innings)?                  |
| Q6  | Which are the top 5 venues where Sachin scored the most runs?                             |
| Q7  | How does Sachin's performance compare between Home and Away matches?                      |
| Q8  | How did Sachin's run tally and batting average change year by year?                       |
| Q9  | How was Sachin most frequently dismissed during his career?                               |
| Q10 | How did Sachin perform at different batting positions (1st to 6th)?                       |
| Q11 | What percentage of Sachin's fifties were converted into centuries?                        |
| Q12 | What is the frequency distribution of scores (Duck / 1-9 / 10-24 / 25-49 / 50-99 / 100+)? |
| Q13 | Which were Sachin's top 3 best years by total runs scored?                                |
| Q14 | How did Sachin's strike rate evolve year by year throughout his career?                   |
| Q15 | How did Sachin perform at Home vs Away across each match format?                          |
| Q16 | What is the cumulative run tally across Sachin's entire career (innings by innings)?      |
| Q17 | What are Sachin's top 10 highest individual scores?                                       |
| Q18 | Against which opponents did Sachin average the most runs per innings (min 10 innings)?    |
| Q19 | Against which opponents was Sachin most consistent (lowest score standard deviation)?     |

## Data Dictionary

| Column Name   | Description                                                                  | Data Type     |
|---------------|------------------------------------------------------------------------------|---------------|
| date          | Date of the innings (parsed from dd-Mon-YYYY format)                         | DateTime      |
| runs          | Runs scored in the innings (cleaned of '*' not-out marker)                   | Integer       |
| mins          | Minutes batted (may be null for early records)                               | Float         |
| bf            | Balls faced in the innings                                                   | Float         |
| fours         | Number of fours hit (4s) — renamed from '4s'                                 | Integer       |
| sixes         | Number of sixes hit (6s) — renamed from '6s'                                 | Integer       |
| sr            | Strike rate as recorded; '-' cleaned to NaN                                  | Float         |
| calculated_sr | Derived strike rate: $(\text{runs} / \text{bf}) \times 100$ , rounded to 2dp | Float         |
| pos           | Batting position (1–11)                                                      | Integer       |
| inns          | Innings number within the match (1 or 2)                                     | Integer       |
| dismissal     | Mode of dismissal — standardised (e.g., 'run out', 'not out')                | String        |
| not_out_flag  | 1 if not out (*), 0 if dismissed                                             | Integer (0/1) |
| ground        | Venue name — used for Home/Away classification                               | String        |
| opponent      | Opposing team — standardised names (e.g., 'west indies')                     | String        |
| match_type    | Format: 'test', 'odi', or 't20i'                                             | String        |
| year          | Year extracted from date column                                              | Integer       |
| month         | Month extracted from date column                                             | Integer       |
| home          | Home (Indian ground) or Away classification                                  | String        |
| career_phase  | Early Career (<1998), Prime Career (1998-2006), Late Career (2007+)          | String        |
| is_50         | Flag: 1 if score between 50-99, else 0                                       | Integer (0/1) |
| is_100        | Flag: 1 if score $\geq 100$ , else 0                                         | Integer (0/1) |
| data_quality  | 'complete' if bf is not null, else 'partial'                                 | String        |

## EDA — Exploratory Data Analysis (Python)

### Data Loading

Libraries imported and dataset loaded into a pandas DataFrame.

```
import pandas as pd
import numpy as np

df = pd.read_csv('sachin_performace.csv')
print('Initial Shape:', df.shape)
```

### Initial Data Exploration

Viewed the first/last rows, checked data types, summary statistics, and missing values.

```
df.head()          # First 5 rows
df.tail()          # Last 5 rows
df.describe(include='all')  # Summary statistics
df.info()          # Data types + non-null counts
df.isnull().sum()   # Missing value counts per column
```

*Why: Early inspection reveals shape, data types, and quality issues before any transformation.*

### Step-by-Step Cleaning Operations

#### 1. Remove Unnamed Columns

*What: Dropped auto-generated index columns (Unnamed: 0) that appear when reading CSVs with default index.*

```
df.drop(columns=['Unnamed: 0'], inplace=True, errors='ignore')
```

#### 2. Standardise Column Names

*What: Converted all column names to lowercase, stripped whitespace, replaced spaces with underscores.*

```
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')
```

#### 3. Clean Text Values

*What: Stripped leading/trailing spaces, replaced underscores with spaces, lowercased all string columns.*

```
for col in df.select_dtypes('object').columns:
    df[col] = df[col].astype(str).str.strip().str.lower()
```

#### 4. Handle Missing Values (Runs / SR)

*What: Replaced '-' placeholder strings with NaN in mins and sr columns.*

```
df['mins'] = df['mins'].replace('-', np.nan)
df['sr'] = df['sr'].replace('-', np.nan)
```

#### 5. Extract Not-Out Flag & Clean Runs

*What: Sachin's not-out innings are marked with '\*'. Flag extracted, then asterisk removed.*

```
df['not_out_flag'] = df['runs'].astype(str).str.contains('*').astype(int)
df['runs'] = pd.to_numeric(df['runs'].str.replace('*', ''), errors='coerce')
```

#### 6. Convert Numeric Types

*What: Converted runs, mins, bf, 4s, 6s, sr, pos, inns to numeric — coercing invalid values to NaN.*

```
for col in ['runs', 'mins', 'bf', '4s', '6s', 'sr', 'pos', 'inns']:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

## 7. Fix Date Format

What: Dates were in 'ddMonYYYY' string format — parsed to proper datetime objects.

```
df['date'] = pd.to_datetime(df['date'], format='%d%b%Y')
```

## 8. Standardise Team Names

What: Replaced abbreviations (wi, aus, nz, sl) with full names. Fixed spacing issues.

```
team_map = {'wi': 'west indies', 'aus': 'australia', 'nz': 'new zealand', 'sl': 'sri lanka'}
df['opponent'] = df['opponent'].replace(team_map)
```

## 9. Standardise Dismissal Values

What: Fixed inconsistent dismissal labels — 'runout' → 'run out', 'hitwicket' → 'hit wicket' etc.

```
df['dismissal'] = df['dismissal'].replace({'runout': 'run out', 'hitwicket': 'hit wicket'})
```

## 10. Remove Invalid Rows

What: Removed rows with negative runs, unrealistic strike rates (>1000), and zero balls faced.

```
df = df[df['runs'] >= 0]
df = df[df['sr'].isna() | df['sr'].between(0, 1000)]
df = df[df['bf'].isna() | (df['bf'] > 0)]
```

## 11. Feature Engineering — Year & Month

What: Extracted year and month from date for time-series analysis.

```
df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month
```

## 12. Home / Away Classification

What: Classified each innings as Home (Indian ground) or Away using a list of known Indian venues.

```
home_grounds = ['delhi', 'kolkata', 'chennai', 'mumbai', 'hyderabad',
                 'bengaluru', 'nagpur', 'pune', 'ahmedabad', 'jaipur']
df['home'] = df['ground'].apply(lambda x:
    'Home' if any(c in str(x).lower() for c in home_grounds) else 'Away')
```

## 13. Match Type Standardisation

What: Mapped raw format codes ('odi', 'test', 'ti') to clean labels ('odi', 'test', 't20i').

```
df['match_type'] = df['match'].map({'odi': 'odi', 'test': 'test', 'ti': 't20i'})
df.drop(columns=['match'], inplace=True)
```

## 14. Career Phase Classification

What: Tagged each row with a career phase: Early (<1998), Prime (1998-2006), Late (2007+).

```
def career_phase(y):
    if y < 1998: return 'Early Career'
    elif y < 2007: return 'Prime Career'
    else: return 'Late Career'
```

## 15. Scoring Flags (is\_50 / is\_100)

What: Binary columns flagging half-centuries (50-99) and centuries (100+) for easy aggregation.

```
df['is_50'] = df['runs'].apply(lambda x: 1 if 50 <= x < 100 else 0)
df['is_100'] = df['runs'].apply(lambda x: 1 if x >= 100 else 0)
```

## 16. Calculated Strike Rate

What: Derived strike rate column as (runs / bf) × 100 to cross-validate recorded SR.

```
df['calculated_sr'] = (df['runs'] / df['bf']) * 100
```

## 17. Fill Boundary Statistics

What: Replaced NaN values in fours and sixes columns with 0 (no boundaries recorded).

```
df['4s'] = df['4s'].fillna(0).astype(int)
df['6s'] = df['6s'].fillna(0).astype(int)
```

## 18. Data Quality Flag

What: Added 'complete' / 'partial' quality label based on whether balls-faced data is available.

```
df['data_quality'] = df['bf'].apply(lambda x: 'complete' if pd.notna(x) else 'partial')
```

## 19. Remove Duplicates & Sort

What: Dropped duplicate rows and sorted chronologically by date and innings number.

```
df.drop_duplicates(inplace=True)
df = df.sort_values(by=['date', 'inns'])
```

## 20. Column Rename & Export

What: Renamed '4s'→'fours' and '6s'→'sixes' for SQL compatibility, then exported to CSV.

```
df.rename(columns={'4s':'fours','6s':'sixes'}, inplace=True)
df.to_csv('sachin_tendulkar_cleaned.csv', index=False)
print('Final Shape:', df.shape)
```

## Data Export to SQL Server

After cleaning, the dataset was exported to Microsoft SQL Server using SQLAlchemy + pyodbc for structured query analysis and Power BI connectivity.

```
from sqlalchemy import create_engine
import urllib

connection_string = urllib.parse.quote_plus(
    'DRIVER={ODBC Driver 17 for SQL Server};'
    'SERVER=JYOTHSNA;DATABASE=sachin_career_analysis_project;'
    'Trusted_Connection=yes;'
)

engine = create_engine(f'mssql+pyodbc:///odbc_connect={connection_string}')
df.to_sql('sachin_career', engine, if_exists='replace', index=False)
print('Data loaded into SQL Server successfully!')
```



## Data Analysis Using SQL Server

All 19 business questions were answered using T-SQL queries on the cleaned sachin\_career table in Microsoft SQL Server. Key techniques used include GROUP BY aggregations, CASE WHEN scoring logic, window functions (SUM OVER), subqueries, TOP N filtering, NULLIF for division safety, and STDEV for consistency analysis.

### Q1 — Total Innings & Runs

```
SELECT COUNT(*) AS total_innings, SUM(runs) AS total_runs
FROM sachin_career;
```

**Result:** 782 innings | 34,357 total runs

### Q2 — Career Batting Average

```
SELECT CAST(SUM(runs)*1.0 / NULLIF((COUNT(*)-SUM(not_out_flag)),0)
          AS DECIMAL(10,2)) AS career_batting_average
FROM sachin_career;
```

**Result:** Career Average: 48.53

### Q3 — Complete Career Summary

```
SELECT COUNT(*) AS innings, SUM(runs) AS total_runs,
       CAST(ROUND(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0),2)
          AS DECIMAL(10,2)) AS career_average,
       SUM(CASE WHEN runs>=100 THEN 1 ELSE 0 END) AS centuries,
       SUM(CASE WHEN runs BETWEEN 50 AND 99 THEN 1 ELSE 0 END) AS fifties,
       MAX(runs) AS highest_score,
       CAST(ROUND(SUM(runs)*100.0/NULLIF(SUM(bf),0),2) AS DECIMAL(10,2)) AS strike_rate
FROM sachin_career;
```

**Result:** Innings:782 | Runs:34,357 | Avg:48.53 | 100s:100 | 50s:164 | HS:248 | SR:67.61

### Q4 — Performance by Format

```
SELECT match_type, COUNT(*) AS innings, SUM(runs) AS total_runs,
       CAST(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0) AS DECIMAL(10,2)) AS avg,
       SUM(CASE WHEN runs>=100 THEN 1 ELSE 0 END) AS centuries,
       MAX(runs) AS highest
FROM sachin_career GROUP BY match_type;
```

**Result:** ODI: 18,426 runs (avg 45) | Test: 15,921 runs (avg 54) | T20I: 10 runs

### Q5 — Top Opponents by Runs

```
SELECT opponent, COUNT(*) AS innings, SUM(runs) AS total_runs,
       CAST(ROUND(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0),2)
          AS DECIMAL(10,2)) AS average, MAX(runs) AS highest
FROM sachin_career GROUP BY opponent
HAVING COUNT(*) >= 10 ORDER BY total_runs DESC;
```

**Result:** Australia: 6,707 runs | Sri Lanka: 5,108 | England: 3,990

### Q6 — Top 5 Venues

```
SELECT TOP 5 ground, COUNT(*) AS innings, SUM(runs) AS total_runs,
    MAX(runs) AS highest
FROM sachin_career
GROUP BY ground ORDER BY total_runs DESC;
```

**Result:** Top ground by runs: Dhaka, Sydney, Ahmedabad, Bengaluru, Colombo

### Q7 — Home vs Away

```
SELECT home, COUNT(*) AS innings, SUM(runs) AS total_runs,
    CAST(ROUND(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0),2)
    AS DECIMAL(10,2)) AS average
FROM sachin_career GROUP BY home;
```

**Result:** Home: 10,810 runs (31.5%) | Away: 23,547 runs (68.5%)

### Q8 — Year-by-Year Progression

```
SELECT year, COUNT(*) AS innings, SUM(runs) AS yearly_runs,
    CAST(ROUND(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0),2)
    AS DECIMAL(10,2)) AS yearly_average
FROM sachin_career
GROUP BY year ORDER BY year;
```

**Result:** Peak: 1998 (2,541 runs) | 2003 (2,201 runs) | 2010 (1,766 runs)

### Q9 — Dismissal Analysis

```
SELECT dismissal, COUNT(*) AS times FROM sachin_career
WHERE not_out_flag=0 AND dismissal!='not out'
GROUP BY dismissal ORDER BY times DESC;
```

**Result:** Caught: 427 | Bowled: 123 | LBW: 102 | Run Out: 43 | Stumped: 12

### Q10 — Performance by Batting Position

```
SELECT pos AS batting_position, COUNT(*) AS innings,
    SUM(runs) AS total_runs,
    CAST(ROUND(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0),2)
    AS DECIMAL(10,2)) AS average
FROM sachin_career GROUP BY pos ORDER BY pos;
```

**Result:** Position 4 (Test anchor role) showed highest average

### Q11 — Fifty-to-Century Conversion

```
SELECT CAST(ROUND(
    SUM(CASE WHEN runs>=100 THEN 1 ELSE 0 END)*100.0 /
    NULLIF(SUM(CASE WHEN runs>=50 THEN 1 ELSE 0 END),0),2)
    AS DECIMAL(10,2)) AS conversion_pct
FROM sachin_career;
```

**Result:** Conversion Rate: ~37.88% (100 centuries from 264 fifties+centuries)

## Q12 — Score Frequency Distribution

```
SELECT score_range, COUNT(*) AS frequency,
       CAST(ROUND(COUNT(*)*100.0/(SELECT COUNT(*) FROM sachin_career),2)
            AS DECIMAL(10,2)) AS percentage
FROM (
  SELECT CASE WHEN runs=0 THEN 'Duck'
             WHEN runs<10 THEN '1-9' WHEN runs<25 THEN '10-24'
             WHEN runs<50 THEN '25-49' WHEN runs<100 THEN '50-99'
             ELSE '100+' END AS score_range FROM sachin_career) t
GROUP BY score_range;
```

**Result:** 100+: 100 inns | 50-99: 164 | 25-49: ~150 | Ducks: ~34

## Q13 — Top 3 Best Years

```
SELECT TOP 3 year, SUM(runs) AS total_runs, COUNT(*) AS innings
FROM sachin_career GROUP BY year ORDER BY total_runs DESC;
```

**Result:** #1: 1998 (2,541) | #2: 2003 (2,201) | #3: 1996 (2,234)

## Q14 — Strike Rate by Year

```
SELECT year,
       ROUND(SUM(runs)*100.0/NULLIF(SUM(bf),0),2) AS strike_rate
FROM sachin_career
WHERE bf IS NOT NULL GROUP BY year ORDER BY year;
```

**Result:** SR increased in later ODI years; Test SR remained steady ~50-55

## Q15 — Home vs Away by Format

```
SELECT home, match_type, COUNT(*) AS innings, SUM(runs) AS total_runs,
       CAST(ROUND(SUM(runs)*1.0/NULLIF(COUNT(*)-SUM(not_out_flag),0),2)
            AS DECIMAL(10,2)) AS average
FROM sachin_career GROUP BY home, match_type ORDER BY match_type;
```

**Result:** Away Test average notably higher — away conditions tested true class

## Q16 — Cumulative Career Runs (Window Function)

```
SELECT CAST(date AS DATE) AS date, runs,
       SUM(runs) OVER (ORDER BY date, inns) AS cumulative_runs
FROM sachin_career ORDER BY date, inns;
```

**Result:** Used for career progression line chart in Power BI (0 → 34,357)

## Q17 — Top 10 Highest Scores

```
SELECT TOP 10 runs, opponent, ground, CAST(date AS DATE) AS date, match_type
FROM sachin_career ORDER BY runs DESC;
```

**Result:** #1: 248 vs Bangladesh, Dhaka | #2: 241 vs Australia, Sydney

## Q18 — Best Batting Average vs Opponents

```
SELECT TOP 5 opponent, COUNT(*) AS innings, SUM(runs) AS total_runs,
       ROUND(AVG(CAST(runs AS FLOAT)),2) AS batting_avg
FROM sachin_career
GROUP BY opponent HAVING COUNT(*)>=10 ORDER BY batting_avg DESC;
```

**Result:** Highest average opponents identified (min 10 innings filter applied)

### Q19 — Consistency Analysis (STDEV)

```
SELECT opponent, ROUND(AVG(CAST(runs AS FLOAT)),2) AS avg_runs,  
       ROUND(STDEV(CAST(runs AS FLOAT)),2) AS consistency  
FROM sachin_career  
GROUP BY opponent HAVING COUNT(*)>=10 ORDER BY avg_runs DESC;
```

**Result:** Lower STDEV = more consistent; higher = streaky performances

Power BI Dashboard

The cleaned SQL Server data was connected to Power BI to build an interactive dashboard titled 'Sachin Tendulkar — Career Performance Dashboard'. The dashboard provides slicers for Match Type, Opponent, Ground, Home/Away, and Year range, enabling dynamic exploration.

Dashboard Visuals & KPI Cards

| Visual                   | Chart Type       | Insight                                                                      |
|--------------------------|------------------|------------------------------------------------------------------------------|
| KPI Cards (6)            | Card Visuals     | Total Runs · Batting Avg · Centuries · Fifties · Highest Score · Strike Rate |
| Runs Over Years          | Line Chart       | Year-by-year run tally — peak at 1998 (2,541 runs)                           |
| Runs by Opponent         | Horizontal Bar   | Australia leads at 6,707 runs; Kenya lowest                                  |
| Home vs Away Split       | Donut Chart      | Away 68.54% (23,547) vs Home 31.46% (10,810)                                 |
| Runs by Match Type       | Clustered Bar    | ODI: 18,426   Test: 15,921   T20I: 10                                        |
| Dismissal Analysis       | Bar Chart        | Caught: 427 most common dismissal                                            |
| Format-Wise Performance  | Combo (Bar+Line) | Total Runs bars + Batting Average line overlay                               |
| Cumulative Career Runs   | Area Chart       | Running total from 0 → 34,357 over career                                    |
| Top 10 Individual Scores | Table            | Ranked innings with opponent, ground, date, format                           |
| Score Range Distribution | Grouped Bar      | Duck/1-9/10-24/25-49/50-99/100+ by format                                    |
| Slicers (Filters)        | Dropdown/Button  | Match Type · Opponent · Ground · Home/Away · Year slider                     |

Dashboard Design Features

- Cricket-themed dark background with Sachin's image integrated as a visual backdrop
- Gold accent colours for KPI values — reflecting the 'God of Cricket' branding
- 24 Years of Glory tagline with 1989–2013 span and innings/not-out counts displayed
- Fully interactive — all charts cross-filter each other via Power BI slicer connectivity
- Two dashboard pages: Overview (career stats) and Detail (top/bottom products by rating & revenue)

## Key Insights & Findings

### Total Career Runs — 34,357

Sachin is the all-time leading run scorer in international cricket. 782 innings across 24 years place him in a class of his own. His closest rivals trail by thousands of runs.

### 100 International Centuries

The only cricketer to score 100 international centuries. The next best is Ricky Ponting with 71. This milestone remains one of sport's most extraordinary achievements.

### Batting Average — 48.53 (Overall)

With not-outs excluded from the denominator, his 48.53 average reflects true batting quality. His Test average of 53.78 puts him among the all-time elite.

### Australia — Favourite Opponent (6,707 runs)

Despite Australia producing legendary fast bowlers (McGrath, Warne, Lee), Sachin scored more runs against them than any other team — demonstrating class against the best.

### Peak Year: 1998 (2,541 runs)

1998 was arguably Sachin's greatest year — he dismantled Shane Warne in India and dominated the Coca-Cola series in Sharjah. His Desert Storm innings remain legendary.

### Away Dominance (68.54% of runs)

Counter-intuitively, 68.54% of Sachin's runs came away from home — proving he excelled on the toughest pitches worldwide, not just on subcontinental turning tracks.

### Caught — Most Common Dismissal (427 times)

Being caught (427 times out of 708 dismissals = ~60%) is the most frequent dismissal, consistent with elite batsmen who attack — more chances, more catches.

### Fifty-to-Century Conversion ~37-38%

Converting roughly 38% of fifties into centuries shows strong mental fortitude. Elite batsmen typically convert 30-40%, and Sachin is firmly in that bracket.

### Strike Rate Evolution

His ODI strike rate increased over the years as the format demanded more aggression, while his Test SR remained steady — showing excellent format adaptability.

### Career Longevity — 24 Years

Playing from age 16 (1989) to 40 (2013) across all formats is a physical and mental achievement unmatched in cricket. The cumulative runs chart shows steady upward growth.

Tools & Technologies Used

| Tool / Technology   | Purpose                   | Key Operations                                                                        |
|---------------------|---------------------------|---------------------------------------------------------------------------------------|
| Python (pandas)     | Data Cleaning & EDA       | Column standardisation, missing value handling, type conversion, feature engineering  |
| Python (numpy)      | Numerical Operations      | NaN handling, numeric coercion, flag creation                                         |
| SQL Server (T-SQL)  | Data Analysis             | 19 business queries — aggregations, window functions, CASE WHEN, STDEV, NULLIF        |
| SQLAlchemy + pyodbc | Python → SQL Export       | Cleaned DataFrame exported to SQL Server table via SQLAlchemy engine                  |
| Power BI Desktop    | Dashboard & Visualisation | KPI cards, line/bar/donut charts, cumulative area chart, slicers, interactive filters |
| ESPN Cricinfo       | Data Source               | Innings-level batting records for all international formats (1989–2013)               |

Conclusion

This project demonstrates a complete, end-to-end data analytics workflow applied to one of the richest sports datasets available — Sachin Tendulkar's 24-year international career.

Starting from raw ESPN Cricinfo data, we performed 20 structured Python cleaning operations — handling missing values, standardising categories, extracting features, and validating data quality. The cleaned data was exported to SQL Server where 19 targeted T-SQL queries answered every business question — from career summaries to format breakdowns, opponent analyses, home/away comparisons, dismissal patterns, and consistency metrics using STDEV.

The Power BI dashboard synthesises all insights into an interactive, cricket-themed visual experience — complete with KPI cards (34,357 runs, 100 centuries, 48.53 average), dynamic charts, and cross-filtering slicers — enabling any cricket fan or analyst to explore the data freely.

The numbers confirm what cricket lovers always knew: Sachin Tendulkar was not just great — he was consistent, adaptable, and relentless across formats, opponents, and geographies for a quarter of a century. The God of Cricket. 1989 – 2013.

| Metric           | Value  |
|------------------|--------|
| Total Innings    | 782    |
| Not Outs         | 74     |
| Total Runs       | 34,357 |
| Career Average   | 48.53  |
| Centuries (100s) | 100    |

|                      |                                      |
|----------------------|--------------------------------------|
| Half-Centuries (50s) | 164                                  |
| Highest Score        | 248 (vs Bangladesh, Dhaka, Dec 2004) |
| Overall Strike Rate  | 67.61                                |
| Career Span          | 15 Nov 1989 – 14 Nov 2013 (24 Years) |
| Formats Played       | Test   ODI   T20I                    |



# THE END